The empirical recurrence for OEIS A251041, recovered from its tail

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Abstract

OEIS A251041 records "Empirical recurrence of order 98" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 557$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 101$. Its last coefficient is $c_{98} = 46080 \neq 0$, so no further tail satisfies anything shorter; any order-98 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A251041 is "Number of (4+1)X(n+1) 0..3 arrays with nondecreasing min(x(i,j),x(i,j-1)) in the i direction and nondecreasing absolute value of x(i,j)-x(i-1,j) in the j direction.". It has offset 1 and begins

$$19112, 304421, 3396593, 32439684, 267606254, 2010619868, 13920865807, 90738782195, \ \ldots$$

The entry states:

> Empirical recurrence of order 98 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 13:02:14, revision 8) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 557$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 557$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 98: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 98.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1114$ exact terms from $n = 3$ on, it returns order 98 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{98} c_i\, a(n-i),$$

c_1	19	c_{26}	949588779776921	c_{51}	22542895355802865055	c_{76}	−604988993487251
c_2	−56	c_{27}	5848338041308553	c_{52}	−63197678523602980325	c_{77}	299986912484709
c_3	−1118	c_{28}	−5263033329376759	c_{53}	−21464075048097899536	c_{78}	1491559407213198
c_4	7787	c_{29}	−22228577231814635	c_{54}	66020854132323812229	c_{79}	−86246108881941
c_5	23338	c_{30}	23797321571839956	c_{55}	18149532499310702050	c_{80}	−31528915640701
c_6	−332725	c_{31}	74386389495689363	c_{56}	−61936439626247645838	c_{81}	200841245184424
c_7	15459	c_{32}	−91089619177385447	c_{57}	−13562954028130676878	c_{82}	565085219066958
c_8	8269181	c_{33}	−220194399907839106	c_{58}	52142531518184415036	c_{83}	−3804973939195
c_9	−12691955	c_{34}	301095054247025748	c_{59}	8893006721254744186	c_{84}	−8468231518758
c_{10}	−140150103	c_{35}	578706693143571205	c_{60}	−39349528986234810290	c_{85}	5825535989468
c_{11}	369508735	c_{36}	−870107876919718899	c_{61}	−5059376414166784738	c_{86}	10422886524856
c_{12}	1734765700	c_{37}	−1354280850583424612	c_{62}	26578405130698503061	c_{87}	−709608540616
c_{13}	−6475894859	c_{38}	2216494312347457308	c_{63}	2450923581735401264	c_{88}	−1029194925232
c_{14}	−16137154367	c_{39}	2828223225512715486	c_{64}	−16036495499760831349	c_{89}	67019223696
c_{15}	82797452927	c_{40}	−5006383592575783372	c_{65}	−974864412068601328	c_{90}	78959406048
c_{16}	112153771998	c_{41}	−5279104545151748917	c_{66}	8622287041679259061	c_{91}	−4712586912
c_{17}	−830360451505	c_{42}	10069211587283309409	c_{67}	290948864052191108	c_{92}	−4494972352
c_{18}	−540207021955	c_{43}	8816268270141575957	c_{68}	−4118865223430968673	c_{93}	230827328
c_{19}	6790250508056	c_{44}	−18090125989824815711	c_{69}	−43715328269559239	c_{94}	176808832
c_{20}	1123778893886	c_{45}	−13178949727396636587	c_{70}	1741888639208094162	c_{95}	−6963840
c_{21}	−46398885778623	c_{46}	29097445213704156317	c_{71}	−15652453141925606	c_{96}	−4237056
c_{22}	9000049492278	c_{47}	17631738502399190471	c_{72}	−649365118521111226	c_{97}	96000
c_{23}	269468333597655	c_{48}	−41969575123006187367	c_{73}	16479328846257223	c_{98}	46080
c_{24}	−128059089399709	c_{49}	−21096505045617910132	c_{74}	212302671866193853		
c_{25}	−1346808035903941	c_{50}	54342020704194763112	c_{75}	−8244615951640067		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 101$, and it is the recurrence OEIS A251041 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{98} - c_1 x^{98-1} - \cdots - c_{98}$. Its constant term is $-c_{98} = -46080$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 98 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 98, so $\deg q = 98$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 98, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 98, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 46080.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-98 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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